

SUSPECT ADVERSE REACTION REPORT
CIOMS FORM I : CASE 2

I. REACTION INFORMATION

1. PATIENT INITIALS (first, last)	1a. COUNTRY	2. DATE OF BIRTH Day Month Year	2a. AGE Years	3. SEX	4-6. REACTION ONSET Day Month Year
Not Reported	Italy	Not Reported	65 Years	Male	UNKNOWN (12 days after initiation)
8-12. CHECK ALL APPROPRIATE TO ADVERSE REACTION [X] PATIENT DIED [X] INVOLVED OR PROLONGED INPATIENT HOSPITALISATION [X] LIFE THREATENING [] INVOLVED PERSISTENCE OR SIGNIFICANT DISABILITY OR INCAPACITY			7 + 13. DESCRIBE REACTION(S) (including relevant tests/lab data) Twelve days after receiving a single cycle of pembrolizumab, the patient presented with fever (37.8°C), severe muscle aches, constipation, diplopia, and gait instability. On hospital admission (Day 15), neurological deficits included impaired lateral rotation of the right eye, left eyelid ptosis, and reduced motor strength in the lower limbs. On day 2 of hospitalization, the patient developed acute palpitations, severe dyspnea, and atrial fibrillation, which progressed on day 3 to a second-degree Mobitz Type 2 atrioventricular block, followed by combined respiratory and renal failure. Despite mechanical ventilation and continuous renal replacement therapy, the patient suffered refractory hypoxemia and sustained ventricular tachycardia with peak torsades de pointes fibrillation, resulting in death 15 days post-infusion. Diagnosis: Pembrolizumab-induced triad of Myasthenia Gravis, Lympho-histiocytic Myocarditis, and Necrotizing Myositis, complicated by acute kidney injury, hepatic injury, respiratory failure, atrioventricular block, and fatal ventricular tachycardia. Lab Values: • Serum creatinine: 2.49 mg/dL • GFR: 23 mL/min/1.73 m² • ALT: 659 U/L AST: 1687 U/L		

	• CRP: 7.9 mg/dL • Troponin I: 25,010 ng/L • Creatine Kinase (CK): 14,977 U/L Post-Mortem / Autopsy Findings: Global cardiomegaly (530 g) with a marbled left ventricular myocardium. Histopathology confirmed severe lympho-histiocytic myocarditis with contraction band necrosis and dense CD3+ and CD8+ cytotoxic T-cell infiltrates (Tia-1+, PD-1+), along with robust focal upregulation of PD-L1 on damaged cardiomyocytes. Iliopsoas muscle biopsy showed severe necrotizing myositis with lymphocytic infiltration (CD3+, CD20+) and focal PD-L1 expression on injured myofibers. Additional findings included diffuse acute tubular necrosis, severe centrilobular hepatic necrosis, and focal acute bronchopneumonia.
--	---

II. SUSPECT DRUG(S) INFORMATION

14. SUSPECT DRUG(S) (include generic name) Pembrolizumab	
15. DAILY DOSE(S) 200 mg (Single cycle administered)	16. ROUTE(S) OF ADMINISTRATION Intravenous
17. INDICATION(S) FOR USE Adjuvant treatment for advanced metastatic nodular melanoma of the left forearm (pT4a pN1b, NRAS-mutated)	
18. THERAPY DATES (from/to) UNKNOWN – RELATIVE CHRONOLOGY USED Start: Approximately 3 months after histological diagnosis Stop: Discontinued after 1 dose due to immediate symptom onset	19. THERAPY DURATION 1 single dose (1 day)
20. DID REACTION ABATE AFTER STOPPING	21. DID REACTION REAPPEAR AFTER

DRUG? ☐ YES ☒ NO ☐ NA Negative Dechallenge: Condition rapidly and continuously deteriorated despite drug withdrawal and intensive medical management.	REINTRODUCTION? ☐ YES ☐ NO ☒ NA Not Performed: Rechallenge was not attempted due to the fatal outcome.

III. CONCOMITANT DRUG(S) AND HISTORY

22. CONCOMITANT DRUG(S) AND DATES OF ADMINISTRATION (exclude those used to treat reaction) None reported for the oncology regimen. (Note: Therapy administered to manage the adverse events included methylprednisolone 60 mg IV QD, pyridostigmine 60 mg PO TID, and anticoagulants).
23. OTHER RELEVANT HISTORY (e.g. diagnostics, allergies, pregnancy with last month of period, etc.) Primary Malignancy: Nodular melanoma of the left forearm (pT4a, Breslow thickness 12 mm, mitoses $\geq$ 10 mm ² , vertical growth phase, positive deep surgical margin; NRAS mutations detected: Q61K, Q61R, Q61L, Q61H); Left axillary lymphadenectomy confirmed 1/4 positive lymph node (pN1b). Cardiovascular History: Coronary artery disease with severe diffuse wall calcification and atherosclerosis (80% luminal occlusion of the left main trunk).

IV. MANUFACTURER INFORMATION

24a. NAME AND ADDRESS OF MANUFACTURER Unknown (Literature report)	24b. MFR CONTROL NO. Unknown (Literature report)	24c. DATE RECEIVED BY MANUFACTURER Unknown (Literature report)
24d. REPORT SOURCE ☐ STUDY ☒ LITERATURE ☒ HEALTH PROFESSIONAL	DATE OF THIS REPORT 11-Jul-2026	25a. REPORT TYPE ☒ INITIAL ☐ FOLLOWUP

Literature Citation Details:

Authors: Elena Giovannini, Maria Paola Bonasoni, Michele D'Aleo, Ione Tamagnini, Matteo Tudini, Paolo Fais, Susi Pelotti

Journal: International Journal of Molecular Sciences (2023)

DOI: <https://doi.org/10.3390/ijms241310919>